Environmental Science & Technology

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Number of Carbons Is a Critical Parameter for Accumulation of Per- and Polyfluoroalkyl Substances in the Human Brain

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Cite This: *Environ. Sci. Technol.* 2025, 59, 3366−3375

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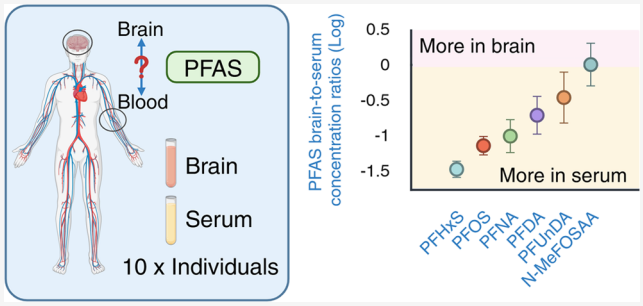
ABSTRACT: Per- and polyfluoroalkyl substances (PFAS), a large group of manmade chemicals, have been detected extensively in the blood of people living in developed countries. Although it has been suggested that PFAS exposure might be associated with harmful effects on the brain, few studies have assessed the presence of PFAS in brain tissues. This study aimed to evaluate the concentrations of a broad range of PFAS in paired postmortem human brain and serum samples and investigate brain-to-serum concentration ratios. A partitioning experiment using PFAS-fortified animal brain samples additionally investigated differences in distribution between lipid-rich brain and water for different PFAS. Out of the 43 PFAS analyzed, 5 were detected in all paired human brain and serum samples, 11 were found in all serum, and 7 were found in all brain samples. Two PFAS compounds were observed at notably higher detection frequencies in brain samples compared to serum. The brain-to-serum ratios of PFAS concentrations ranged from approximately 0.04 for perfluorohexanesulfonate (PFHxS) to 1.3 for *N*-methyl perfluorooctanesulfonamido acetic acid (*N*-MeFOSAA) with a clear increase in PFAS brain-to-serum ratios with the total number of carbons. There were no differences between the two cortical brain regions analyzed. Results underscore the necessity of a better understanding of individual PFAS, as the difference in their properties can influence their behavior within the human brain.



KEYWORDS: *per- and polyfluoroalkyl substances, PFAS, brain, serum, accumulation, carbon, partition*

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1. INTRODUCTION

Per- and polyfluoroalkyl substances (PFAS) are a group of manmade chemicals. PFAS possess useful properties, including both water and oil repellence and resistance to high temperatures, resulting in their multiple industrial applications and consumer products such as food packaging, furniture, firefighting foam, and cookware.[1,2] Humans are exposed to PFAS through multiple pathways, including via drinking water, consumption of contaminated food, indoor dust ingestion and inhalation, and the use of PFAS-contaminated products, leading to a concern about health risks.[3,4] Potential health effects of PFAS that have been reported include immunosuppression in early childhood, thyroid disease, increased cholesterol levels, liver damage, decreased kidney function, testicular cancer, and lower birth weight.[5−9] Some studies have reported associations between PFAS exposure and neurocognitive function. For instance, Kawabata et al.[2] found that perfluorododecanoic acid (PFDoDA) can induce cognitive-deficit in adult rats, while Lenters et al.[10] have observed a correlation between early life exposure to perfluorooctanesulfonate (PFOS) and increased risk of attention-deficit/hyperactivity disorder (ADHD) in school-aged children, in which a stronger association was observed for girls.

A wide range of PFAS has been detected in human samples. Nowadays, it is hard to find individuals in developed countries whose blood samples do not have PFAS at measurable levels.[5] Although research on PFAS accumulation in the human body has focused on blood and breast milk samples, an increasing number of studies over the past few decades have explored accumulation in solid tissues. After PFAS enter the human body, they are distributed via the bloodstream and accumulate in the liver, kidney, and spleen and they have also been detected in human brain samples.[11−13] The brain is protected from chemicals that are not needed for its function by the blood−brain barrier (BBB).[14] Nevertheless, studies have detected PFAS in brain tissues and/or cerebrospinal fluid (CSF) in animals and humans.[11,12,15,16] Pérez et al.[12] measured PFAS in human blood and organs, including the brain, and reported several PFAS to be present in the brain tissue. Wang et al.[11] analyzed 223 paired serum and CSF samples, detecting

Received: September 5, 2024
Revised: January 30, 2025
Accepted: January 30, 2025
Published: February 10, 2025

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six PFAS in more than half the CSF samples. Greaves et al.[17] investigated the distribution of PFAS in various regions of the brain of polar bears, showing that PFAS concentrations were brain region-specific. These studies suggest that PFAS are potentially accumulating in brain tissues. However, one of the issues in analyzing chemicals in the brain is the uncertainty related to residual blood in brain tissues. It is currently not clear if PFAS that have been detected in human brain samples in previous studies are attributable to the remaining blood that resides in the postmortem brain tissue. One recent study has attempted to address this issue through the measurement of PFAS in paired plasma and brain tissues from glioma patients, reporting that lower-molecular-weight PFAS preferentially accumulated in brain tissues.[18] However, only a small number of brains have been studied (six paired brain tissue and plasma samples), and only a few PFAS were detected/reported. Inconsistencies in those findings require replication and investigation, including the potential contribution of PFAS from residual blood within brain tissue and potential differences in distribution between blood and brain compartments across different PFAS.

This study investigated the presence and concentrations of PFAS in the human brain and serum, and aimed to provide reliable estimates of brain-to-serum concentration ratios for a broad range of PFAS, and assess the potential influence of residual blood on the measurement of PFAS in brain tissues. Additionally, a partitioning experiment using PFAS-fortified animal brain samples was conducted to explore differences in distribution between the lipid-rich brain and water across different PFAS. The result of the present study enhances our understanding of PFAS accumulation in the human brain, which is important for evaluating potential health effects, including those related to neurodegenerative diseases.

2. MATERIALS AND METHODS

The present study consists of three components, (1) hemoglobin (Hb) measurement in the brain as surrogate measure for the relative contribution of blood in the studied brain tissues, (2) measurement of PFAS in paired human serum and brain samples, and (3) partition experiments to investigate the affinity of different PFAS to a brain (lipid-rich) or aqueous phase.

2.1. Sample Collection. Paired human serum and postmortem brain samples were obtained from 10 individuals for this study. Details of each sample, including age at death, postmortem delay (PMD), and time of serum collection, are shown in Table 1. Individuals were recruited from FRONTIER, the frontotemporal dementia clinical research group at the Brain and Mind Centre, the University of Sydney. Informed consent was obtained from the participant and/or next of kin/primary carer. Ethical approval for blood and brain donation was granted by South Eastern Sydney Health District Ethics Committee HREC 10/092. Project-specific ethical approval was granted by the University of New South Wales Research Ethics Committee A and The University of Queensland Research Ethics and Integrity (HC20189).

Blood samples were collected in 9 mL tubes (BD Vacutainer SST II Advance Tube #367958) and centrifuged at 247*g* for 10 min at 4 °C to obtain serum. Serum samples were stored at −80 °C until used. Brain tissue was collected at limited brain-only autopsy by the Sydney Brain Bank under ethical approval from the University of New South Wales (HC200026) and half of the brain was sliced coronally fresh and stored frozen at −80 °C until used. Cortical tissue samples from the temporal pole and middle frontal were examined to see if there was a difference in PFAS concentrations between these different brain regions, as a previous study suggested region-specific differences in PFAS exist in the brains of polar bears.[17]

Table 1. Sex, Age of Death, Postmortem Delay (PMD), and Number of Weeks before Death When Serum Collected

sample number	sex	age of death	PMD (h)	serum collection (weeks before death)
1	female	72	29	74
2	female	72	25	18
3	male	56	22	33
4	male	73	3	31
5	female	72	10	57
6	female	75	14	1
7	female	79	17	9
8	female	64	18	119
9	male	66	25	31
10	male	69	19	179

2.2. Hemoglobin Quantification. Concentrations of hemoglobin (Hb) in brain postmortem samples were measured to determine the amount of residual blood in these brain samples. The mass of Hb in the brain was measured by retrieving 1 g of a brain sample followed by homogenization in 1.5 mL of phosphate-buffered saline. Samples were then sonicated (3 × 10 s on/off) on ice before centrifuging at 13,000*g* for 30 min at 4 °C. The supernatant was collected and stored at −20 °C. Brain hemoglobin in the supernatant was measured using a hemoglobin assay kit (MAK115-1KT, Sigma-Aldrich) according to the manufacturer's protocol.

2.3. Sample Processing for PFAS Analysis. Details including chemicals, native and internal standards (IS), and instruments used for the extraction are explained in the Supporting Information (SI, Table S1). Samples were analyzed for 43 PFAS. Extraction and analyses were performed according to the methodology described in previous studies.[19,20]

2.3.1. Extraction of Serum Samples. Serum samples (200 μL) were spiked with 2 ng of mass-labeled IS. Serum was extracted with 1.5 mL of acetonitrile by vortex mixing and ultrasonication for 30 min. Samples were then centrifuged at 6950*g* for 30 min. The supernatant was filtered through syringe filters (Phenomenex, 0.2 μm RC membrane) and collected in a 1.5 mL plastic vial. The filtered samples were concentrated to approximately 200 μL under a gentle nitrogen stream and made up to 500 μL by the addition of RO water. Finally, 2 ng of mass-labeled recovery standard was added.

2.3.2. Extraction of Brain Samples. All brain samples were homogenized using a hand-held homogenizer (POLYTRON PT1200 from Kinematica). Before extraction, 2 ng of IS was spiked to approximately 200 mg of brain samples. Sodium hydroxide (200 mM in 400 mL of methanol) was used to digest the brains. The samples were vortex-mixed, sonicated with acetonitrile, and centrifuged at 1450*g*. The supernatant was transferred into a fresh 15 mL falcon tube. The leftover brain slurry was subjected to a second round of extraction with acetonitrile, and the supernatants were combined into one. The supernatant was concentrated to 2 mL by using a gentle stream of nitrogen. Cleanup was done by vortexing samples for 30 s with 1 mL of *n*-hexane. The acetonitrile phase was blown

https://doi.org/10.1021/acs.est.4c09458
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down to 250 μL with nitrogen and topped up to 1 mL with MeOH. The extract was passed through a carbon cartridge (Bond Elut, 100 mg, Agilent Technologies) for a final cleanup. Samples were evaporated to 200 μL under a gentle stream of nitrogen, and the final volume was adjusted to 500 μL by adding Milli-Q water. Finally, samples were spiked with 2 ng of the recovery standard mix.

2.3.3. Partition Experiment. Homogenized goat and sheep brains, each in triplicate, were used for the partition experiments. Approximately 0.5 g of brains were weighed, transferred to a 15 mL falcon tube, and spiked with 8 ng of native PFAS standard mix containing 43 PFAS including 13 perfluoroalkyl carboxylic acid (PFCAs), 9 perfluoroalkane sulfonates (PFSAs), 4 fluorotelomer sulfonates (FTSs), 5 perfluoroalkane sulfonamides (FASAs), and 4 fluorotelomer carboxylates (FTCAs). RO water (2 mL) was added, and samples were vortex-mixed and sonicated for 30 min, followed by centrifuging for 30 min at 5250*g*. The supernatant was collected in a fresh Eppendorf tube and centrifuged again for 30 min at 15,210*g*. The supernatant was filtered through a 0.2 μm RC membrane and collected into a 1.5 mL plastic vial. A total of 300 μL of samples from each sample were transferred to a new 1.5 mL vial, respectively, and 190 μL of MeOH were added. Finally, samples were spiked with 2 ng IS and 2 ng recovery standard mix.

2.4. Instrument Analysis. The human brain and serum PFAS analyses were carried out on high-performance liquid chromatography (HPLC, Nexera, Shimadzu Corp., Kyoto Japan), coupled to tandem mass spectrometry (SCIEX Triple Quad 7500 Concord, Ontario, Canada). Sample analysis for the partition experiment was carried out on a SCIEX Triple Quad 6500+. For both instruments, the mass spectrometers were equipped with an electrospray ionization source and run on negative ionization mode in scheduled multiple reaction monitoring mode (MRM). A Kinetex EVO C18 column (2.6 μm, 100 Å, 100 mm × 2.1 mm, Phenomenex) was utilized to separate analytes. The HPLC was operated on a gradient elution using 1% (aqueous phase A) and 90% (Organic Phase B) MeOH in RO water with 2 mM ammonium acetate as mobile phases. A series of calibration standards ranging from 0.001 to 25 ng/mL (Triple Quad 7500), and 0.01 to 100 μg/L (Triple Quad 6500+) were run before and after sample analysis. In addition, continuing calibration verification standards (CCV) of 1 and 10 μg/L were added between every 12 and 14 samples to observe potential instrumental variations. Both the linear and branched isomers of PFOS were quantified, whereas only the linear isomers of other PFAS are reported.

2.5. Quality Assurance and Quality Control. Details for the quality assurance and quality control (QAQC) results are found in the SI (Tables S2−S4). For serum sample analysis, two blanks (acetonitrile and RO water), one serum duplicate, an in-house reference sample (pooled serum samples with known PFAS serum concentrations from previous analysis), QAQC samples (both unspiked and spiked with native standard mix), and a standard reference material of human serum (SRM, NIST 1957 from the National Institute of Standards and Technology) were extracted together with samples. Additionally, two nonextracted side spikes (NESS, spiked with PFAS native standard mix) were prepared and analyzed. The mean percentage difference of duplicate samples ranged from 0.4 to 39% across different PFAS. The mean percentage difference of the NIST 1957 PFAS concentrations varied from 2.2 to 50% compared to reference values. Native spike recovery was calculated by comparing the QAQC reference serum sample that was spiked with native PFAS standard before extraction, with a QAQC reference serum sample that was spiked after extraction. Average recoveries of IS and native standards ranged from 69 to 132% and from 72 to 122%, respectively.

The brain samples were analyzed in two batches. In each batch, two blanks (acetonitrile and RO water), two duplicate brain samples, and in-house reference samples (fish and prawn tissues whose concentrations are known from a previous analysis, both unspiked and spiked with native standard mix) were included and extracted together with brain samples. NESS samples were also prepared and analyzed alongside each batch. The mean percentage difference of duplicate samples ranged from 5.7 to 49%. Native spike recovery was assessed by comparing reference goat brain samples that were spiked with native PFAS standard mix before and after extraction. The average recovery of mass-labeled IS and native standards ranged from 59 to 137% and from 77 to 136%, respectively.

For partition experiments, two blanks (RO water), control samples (Ctrl, RO water spiked with 8 ng PFAS native standard mix without any brain), were analyzed in the same way as the samples to consider the potential loss of PFAS during the procedure. Two goat brain tissues and two sheep brain tissues (unspiked) were analyzed to assess the PFAS that were present in these brain tissues prior to spiking with native standards. NESS sample (spiked with 2 ng of PFAS native standard mix) was also prepared and analyzed beside samples. The average recovery of mass-labeled IS ranged from 61 to 158%.

The limit of quantification (LOQ) was obtained from a previous study,[19] where instrumental LOQs were determined through the injection of the lowest calibration standard (0.05 μg/L) seven times, and defined as 10 times the standard deviation of these repeated measurements. When compounds were detected at concentrations higher than their LOQs in laboratory blanks included in the extraction procedure, the LOQ was set as the average concentration in these blanks plus three times their standard deviation. LOQ ranges for different PFAS were from 0.001 to 0.24 ng/mL for serum and from 0.002 to 0.30 ng/g for the brain.

2.6. Data Processing and Statistical Analysis. GraphPad Prism 9.5.1 was used for statistical analysis. Pearson correlation coefficients were used to determine if there was a correlation between the amount of Hb in the temporal pole and middle frontal and any relationship between PFAS measurements and the period of time between blood and brain collection. Wilcoxon signed-rank test (nonparametric test) was run to examine statistical differences in PFAS concentrations between temporal pole and middle frontal to clarify if the compounds are present at the same level in the two brain regions. Wilcoxon signed-rank test was also used to compare differences in Hb levels between the temporal pole and the middle frontal. PFAS that were detected in more than 80% of samples was further used to calculate the brain-to-serum ratios of PFAS concentrations. Pearson correlation coefficients were used to evaluate whether the brain-to-serum ratios of PFAS concentrations are correlated with the number of carbons in their structure.

2.6.1. Estimation of the Blood Fraction in the Brain and Blood-Derived PFAS in the Brain. The blood volume per gram of brain tissue was estimated by measuring the Hb

https://doi.org/10.1021/acs.est.4c09458
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Table 2. PFAS Detected in at Least One Sample, Detection Frequency, Mean Concentration, Range of Concentration, Mean and Range of C_{brain}/C_{serum} and C_{brain}/C_{blood}, and Percentage of Blood-Derived PFAS Detected in the Brain

	serum (ng/mL) (N = 10)			brain (ng/g) (N = 10)								
				temporal pole			middle frontal					
PFAS	DF %[a]	mean	range[b]	DF %	mean	range	DF %	mean	range	mean C_{brain}/C_{serum} (range)[c]	mean C_{brain}/C_{blood} (range)[d,e]	blood-derived PFAS in brain (%)
PFBA	60	0.04	0.03−0.07	50	0.15	0.04−0.21	60	0.16	0.04−0.20	N/A	N/A	N/A
PFHxA	10	0.07	0.04−0.07	100	0.08	0.06−0.12	100	0.09	0.06−0.11	N/A	N/A	N/A
PFOA	100	3.5	0.98−6.4	0	<0.15	<0.15	0	<0.15	<0.15	N/A	N/A	N/A
PFNA	100	0.75	0.38−2.0	100	0.07	0.05−0.12	100	0.06	0.04−0.11	0.12 (0.03−0.38)	0.23 (0.05−0.75)	15
PFDA	100	0.27	0.10−0.84	100	0.04	0.03−0.06	100	0.04	0.04−0.07	0.27 (0.06−1.4)	0.50 (0.12−2.8)	7
PFUnDA	100	0.20	0.04−0.98	90	0.04	0.02−0.07	100	0.04	0.04−0.07	0.47 (0.04−1.9)	0.94 (0.08−3.8)	5
PFDoDA	100	0.03	0.02−0.10	10	0.06	0.02−0.06	0	<0.02	<0.02	N/A	N/A	N/A
PFTriDA	10	0.11	0.10−0.11	20	0.15	0.13−0.15	40	0.17	0.13−0.25	N/A	N/A	N/A
PFTeDA	0	<0.03	<0.03	50	0.08	0.04−0.10	50	0.06	0.04−0.10	N/A	N/A	N/A
PFBS	40	0.03	0.008−0.06	0	<0.002	<0.002	0	<0.002	<0.002	N/A	N/A	N/A
PFPeS	100	0.03	0.01−0.05	0	<0.002	<0.002	0	<0.002	<0.002	N/A	N/A	N/A
PFHxS	100	2.4	0.88−3.9	100	0.07	0.05−0.93	100	0.07	0.05−0.97	0.04 (0.02−0.06)	0.07 (0.05−0.11)	37
PFHpS	100	0.34	0.13−0.61	50	0.01	0.003−0.01	50	0.01	0.003−0.01	N/A	N/A	N/A
PFOS	100	6.8	2.4−13	100	0.64	0.28−1.3	100	0.66	0.25−1.6	0.10 (0.06−0.18)	0.21 (0.11−0.36)	12
PFOS total	100	10	3.8−18	100	0.70	0.29−1.4	100	0.70	0.2−1.7	0.08 (0.04−0.14)	0.15 (0.08−0.29)	17
FOSA	0	<0.02	<0.02	100	0.008	0.005−0.009	100	0.07	0.004−0.009	N/A	N/A	N/A
N-MeFOSAA	100	0.11	0.014−0.26	100	0.07	0.007−0.16	100	0.10	0.01−0.23	1.3 (0.19−4.3)	2.6 (0.38−8.6)	1
N-EtFOSAA	70	0.02	0.002−0.059	40	0.02	0.006−0.03	60	0.02	0.006−0.03	N/A	N/A	N/A
9Cl−F53B	100	0.03	0.01−0.13	20	0.02	0.002−0.02	20	0.02	0.002−0.04	N/A	N/A	N/A
∑PFAS		18	6.6−30		1.5	0.86−2.8		1.5	0.69−2.7			

[a]DF: Detection frequency. [b]When a detection frequency is less than 100%, LOQ is used as the lowest end of the range. [c]Concentrations in the brain divided by the concentrations in the serum. [d]Concentrations in the brain divided by the concentrations in the blood. [e]$C_{brain}/C_{blood} = C_{brain}/0.5\ C_{serum}$.

https://doi.org/10.1021/acs.est.4c09458
Environ. Sci. Technol. 2025, 59, 3366−3375

concentration in the temporal pole and middle frontal. It was assumed that Hb in the brain is due to the remaining blood. As Hb is the protein contained in red blood cells, not in serum, the Hb measurement could not be carried out for serum samples. Instead, a reference value of Hb levels in whole blood, 14−18 g/dL for males and 12−16 g/dL for females (average: 16 g/dL for males, 14 g/dL for females)[21] was used. The equation used to calculate the amount of remaining blood in the brain tissue samples is as follows

$$\text{the blood volume per gram of brain (mL/g)} = (\text{Hb in the brain (mg/g)}/1000) / (\text{Hb in the blood (g/dL)}/100) \quad \text{(E1)}$$

To examine the contribution of PFAS derived from blood to PFAS detection in brain samples, further calculation was done. First, PFAS concentrations in the brain (C_{brain}) relative to the whole blood (C_{blood}) were calculated assuming that the concentration of PFAS in whole blood is half the concentrations in serum (eq 2), as it has been shown that the serum-to-whole blood ratios of well-known PFAS are approximately 2.[22,23]

$$C_{brain}\ (\text{ng/g})/C_{blood}\ (\text{ng/mL}) = C_{brain}/0.5 \times C_{serum}\ (C_b/0.5\ C_s) \quad \text{(E2)}$$

Finally, the fraction of PFAS in the brain derived from the blood in the brain was estimated from eq 3.

$$\text{PFAS in brain that is derived from blood (\%)} = (\text{blood volume per gram of brain (mL/g)} \times C_{blood}\ (\text{ng/mL})/C_{brain}\ (\text{ng/g})) \times 100 \quad \text{(E3)}$$

2.6.2. Partition Experiment. PFAS concentrations in the water phase (C_w) and control samples (C_{ctrl}) were measured. The concentrations of PFAS in the brain phase ($C_{b\text{-}p}$, the concentrations in the brain phase in the partition experiments) were calculated by deducting C_w from the concentrations of PFAS in control samples (eq 4). The aim of these experiments was solely to determine the brain-to-water ratios of PFAS concentrations and assess whether those ratios correspond to the brain-to-serum concentration ratios observed in human sample analysis. It is important to note that these partition experiments were not intended to provide a formal partition coefficient, which requires strict definitions and controlled conditions with consideration of various factors, such as chemical equilibrium, sorption to different phases, and pH of the solution.

$$C_{b\text{-}p} = C_{ctrl} - C_w \quad \text{(E4)}$$

3. RESULTS AND DISCUSSION

3.1. PFAS in Brain and Serum. *3.1.1. PFAS Detection.* Overall, up to 18 PFAS compounds out of 43 PFAS analyzed were detected above LOQs in at least one serum or brain sample. Among these, 11 PFAS were found in serum samples of all individuals, while seven PFAS were detected in brain samples of all individuals. Five PFAS were detected in every single individual's sample analyzed. The detection frequency, mean, and ranges of concentration are shown in Table 2.

For serum samples, compounds detected in all individuals include perfluorooctanoic acid (PFOA), perfluorononanoic acid (PFNA), perfluorodecanoic acid (PFDA), perfluoroundecanoic acid (PFUnDA), PFDoDA, perfluoropentanesulfonate (PFPeS), perfluorohexanesulfonate (PFHxS), perfluoroheptanesulfonate (PFHpS), PFOS, *N*-methyl perfluorooctanesulfonamido acetic acid (*N*-MeFOSAA), and 9-chlorohexanadecafluoro-oxanonanesulfonate (9Cl-F53B). Additionally, *N*-ethyl perfluorooctanesulfonamido acetic acid (*N*-EtFOSAA) was detected in 7 out of 10 individuals. PFOS (PFOS total) showed the highest average serum concentration at 6.8 ng/mL, followed by PFOA with an average of 3.5 ng/mL. PFOS accounted for 55% of the total PFAS concentration (∑PFAS) detected in serum samples. The total PFAS concentration ranged from 6.6 to 30 ng/mL across different cases. In brain samples, compounds consistently detected in all individuals included perfluorohexanoic acid (PFHxA), PFNA, PFDA, PFHxS, PFOS, fluorooctane sulfonamide (FOSA), and *N*-MeFOSAA. PFUnDA was detected in 9 out of 10 cases. PFOS (PFOS total) had the highest average concentration at 0.65 ng/g and contributed to about 50% of the total PFAS detected in brain samples. ∑PFAS ranged from 0.69 to 2.7 ng/g.

Some PFAS, including PFHxA, PFTriDA, and FOSA, were either exclusively detected in the brain samples or exhibited significantly higher detection frequencies in the brain compared to in serum. Particularly, PFHxA was found in all of the brain samples but detected in only one serum sample. This result aligns partially with the study by Pérez et al.,[12] which reported a high detection frequency and elevated concentrations of PFHxA in brain tissues. In their analysis of 20 brain samples, PFHxA was predominantly detected at concentrations significantly higher than those of other compounds in the brain.[12] Although serum PFAS levels were not analyzed in their study, the higher detection frequency of PFHxA in the brain is consistent with our findings. However, the extremely elevated concentration of PFHxA relative to other compounds, as reported by Pérez et al.,[12] was not observed in the present study. FOSA also demonstrated a significantly higher detection frequency in the brain compared to that in the serum, being present in all brain samples while remaining undetected in the serum. Pickard et al.[24] suggested that perfluoroalkyl sulfonamides possess a greater propensity for bioaccumulation compared to their terminal degradation products. This property of perfluoroalkyl sulfonamides might explain the consistent detection of FOSA in brain samples but not in serum in the current study.

3.1.2. Comparison between the Temporal Pole and Middle Frontal. Wilcoxon signed-rank tests showed that there was no statistically significant difference in PFAS concentrations between the temporal pole and middle frontal for all of the compounds tested (p value: 0.17 to >0.99) except for *N*-MeFOSAA, which was present at 1.4 times higher average concentration in the middle frontal compared to the temporal pole ($p < 0.05$). Pearson correlation coefficient analysis showed a positive correlation between *N*-MeFOSAA concentrations in the temporal pole and middle frontal (95% confidence interval (CI): 0.81−0.99, $r = 0.95$, $p = 0.0001$).

3.1.3. PFAS Concentrations in Brain Compared to Serum. The brain-to-serum ratios of PFAS concentrations (C_b/C_s) were calculated for the major PFAS compounds that were found in more than 80% of both serum and brain (PFHxS, PFNA, PFOS, PFDA, PFUnDA, and *N*-MeFOSAA) (Table 2 and Figure 1). These ratios represent the PFAS concentrations in the brain (C_b, PFAS concentrations purely measured in the brain tissue samples) relative to those in the serum (C_s). As

https://doi.org/10.1021/acs.est.4c09458
Environ. Sci. Technol. 2025, 59, 3366−3375

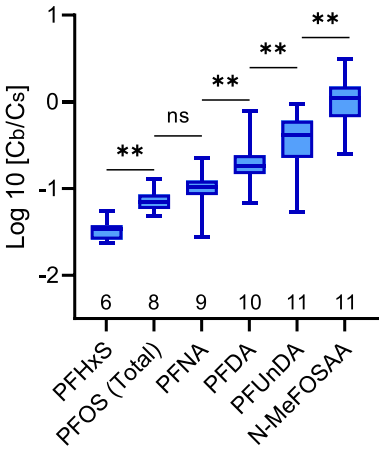

Figure 1. PFAS concentrations in the brain relative to those in blood serum. C_s: PFAS concentrations in the serum. C_b: PFAS concentrations in the brain. Error bars show 5−95 percentile. **: $p < 0.01$, ns: no significance. The numbers along the *x*-axis indicate the number of carbons each compound has (i.f., 6 means the compound has six carbon atoms in its structure).

there was no statistically significant difference in the C_b/C_s value between the temporal pole and middle frontal for all of the compounds (p value: 0.24 to >0.99) except for *N*-MeFOSAA ($p < 0.01$), the ratio C_b/C_s is shown as an average of temporal pole and middle frontal in Table 2 and Figure 1. The C_b/C_s ratios (ratio ± SD) ranged from 0.04 ± 0.01 for PFHxS to 1.3 ± 0.96 for *N*-MeFOSAA (Table 2). C_b/C_s ratios were relatively consistent for any given PFAS, detected in both the serum and the brain samples, across the 10 individuals, but significantly varied between different PFAS (p value: 0.002−0.01). For example, when comparing the C_b/C_s for two PFAS compounds whose number of carbons differed only by one, a significant difference was observed between PFHxS (number of carbons (C) 6) and PFOS (C8) ($p < 0.01$), PFNA (C9) and PFDA (C10) ($p < 0.01$), and PFDA (C10) and PFUnDA (C11) ($p < 0.01$).

The large differences in C_b/C_s ratios among various PFAS clearly indicate that PFAS measured in the brain (i.e., C_b) is not due to PFAS in residual blood but the result of PFAS partitioning across the blood−brain barrier into the brain tissue. Furthermore, a clear increase in the C_b/C_s ratios with an increase in the number of carbons was observed, suggesting that PFAS partitioning into the brain, which is very rich in lipids compared to blood, correlates with the number of carbons in their molecules. Pearson correlation coefficient analysis showed that there is a positive correlation between the C_b/C_s ratios and the number of carbons (95% CI: 0.007−0.99, $r = 0.88$, $p < 0.05$). This implies that PFAS with more carbons have a greater propensity to partition into lipid-rich brain tissue. This is most obvious for some of PFAS with higher number of carbons (i.e., PFTeDA) which were only detected in brain tissues and not in serum. Our findings are in disagreement with the recent study by Xie et al.,[18] which proposed that low-molecular-weight PFAS preferentially accumulate in the brain. Xie et al.[18] reported results from six paired brain tissue and plasma samples, detecting only two compounds, PFOA and PFOS, across all samples, while detection of other analyzed PFAS was variable. In contrast, our study detected a broader spectrum of PFAS with higher detection frequencies for each compound. The comprehensive data provide clearer evidence that PFAS with a higher total number of carbons has a higher tendency to partition into brain tissues compared to PFAS with a lower number of carbons. The results of this study align with a comprehensive review paper on PFAS absorption, distribution, and toxicity in the brain by Cao and Ng.[25] This review paper examined studies that investigated the PFAS brain-to-blood (serum) ratios in animal samples and concluded that the ratios increase with carbon-chain length,[25] which is consistent with our findings in terms of increasing trend in PFAS brain-to-blood (serum) ratios as the number of carbon atoms increases.

N-MeFOSAA, which has 11 carbons, was an exception of the trend observed. As shown in Figure 1, the C_b/C_s ratio for *N*-MeFOSAA was significantly different from the ratio for PFUnDA ($p < 0.01$) even though they have the same total number of carbons. This suggests that, in addition to the number of carbons, there may be some physiochemical properties that play a critical role in determining the distribution of certain PFAS between the brain and blood. As previously mentioned, Pickard et al.[24] suggested that perfluoroalkyl sulfonamides have a higher potential for bioaccumulation compared to their terminal degradation products. This could explain the higher brain-to-serum ratio observed for *N*-MeFOSAA in the present study. Additionally, it is also worth mentioning that different PFAS may have different distribution within the various components of blood, with relevance to the blood matrix tested.[23] In this study, we used a serum-to-whole blood ratio of 2 to estimate blood concentrations based on assessments of several well-known PFAS, which are understood to be almost exclusively carried in the blood serum.[22,23] We acknowledge that this brings some uncertainty to our C_b/C_s estimates. To our knowledge, the partition of *N*-MeFOSAA between blood matrices has not been reported previously, and it is possible that *N*-MeFOSAA is associated with the red blood cell fraction in blood, which is separated out during the blood clotting procedure for serum collection, and the measured concentrations of *N*-MeFOSAA in the brain was affected by the blood contamination. Therefore, the C_b/C_s ratio of *N*-MeFOSAA may not reflect the partitioning of *N*-MeFOSAA between the brain and blood. This is the subject of future investigations.

Another limitation of this study may arise from the interval between serum and brain sample collections, as despite long half-lives of PFAS, the concentrations of some of the PFAS, especially shorter-chain PFAS that have a shorter half-life, may decrease in storage. Furthermore, PFAS might be eliminated in the body in vivo from the brain tissue between when the blood was collected and the time of death. However, our analysis showed no correlation between serum PFAS concentrations, brain PFAS concentrations, and collection date, indicating that the interval between serum and brain sample collections may not be a dominant confounding variable in our study.

3.1.4. Estimation of Contribution of Blood-Derived PFAS in the Brain Using Hemoglobin Data. The results for the measurement of Hb concentrations in the brain samples are shown in Table S5. Mean brain Hb concentrations were 3.9 ± 1.2 and 3.3 ± 1.2 mg/g in the temporal pole and middle frontal, respectively. There was no systematic correlation between the temporal pole and middle frontal for Hb concentrations.

Using the Hb measurement in brain samples and assuming that Hb in brain samples is due to remaining blood in the brain, the fraction of PFAS in brain samples was estimated.

https://doi.org/10.1021/acs.est.4c09458
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First, the quantity of blood in brain tissue was calculated from eq 1 using the Hb measurement in the brain and reference values for Hb in whole blood. The results indicated that the remaining blood in the brain ranged from 2.1 to 5.3% (mean 2.7 ± 0.9%) of the volume of the brain tissue samples in temporal pole and 1.4−3.9% (average 2.2 ± 0.7%) in middle frontal. Hb measurement and calculation results are shown in the Supporting Information (Table S5).

Second, the PFAS concentration in the brain relative to blood (C_{brain}/C_{blood}) was calculated using eq 2. The calculation results (means and ranges) are shown in Table 2. Looking at the average of temporal pole and middle frontal, the C_{brain}/C_{blood} varied among compounds from the lowest recorded for PFHxS with 0.07 to the highest for *N*-MeFOSAA with 2.6. Finally, the fraction of PFAS in the brain that might be explained by the blood in the brain was estimated from eq 3 (Table 2). The results suggest that PFAS in the remaining blood in the brain samples may have contributed to an average of 37% of PFHxS measured in the brain but only 1% of *N*-MeFOSAA. It is clear that the PFAS detected in the brain cannot be explained solely by blood contamination, indicating that PFAS are present in the human brain tissue.

3.2. Partition Experiment for PFAS. The results of the brain−water partition experiments using PFAS-spiked goat and sheep brain samples are presented in Figure 2. C_{ctrl}, C_w, and C_b

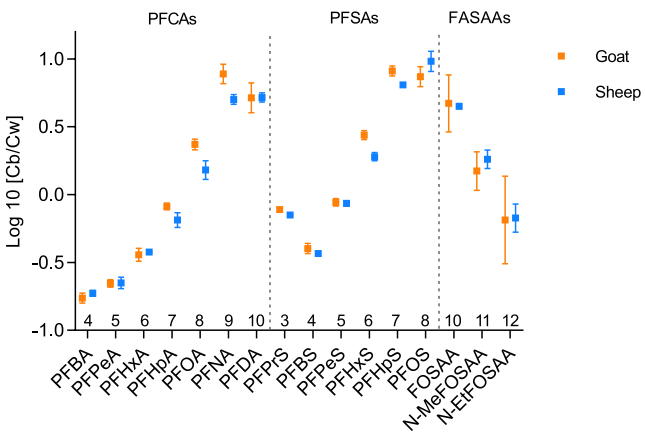

Figure 2. PFAS partition experiment: PFAS concentrations in the brain (C_b) relative to the water (C_w). Compounds are sorted by increasing the number of carbons along the *x*-axis. Error bars show the minimum and maximum values. PFCAs: perfluoroalkyl carboxylates, PFSAs: perfluoroalkane sulfonates, FASAA: perfluoroalkane sulfonamido acetic acids. The numbers along the *x*-axis indicate the number of carbons each compound has (i.f., 6 means the compound has six carbons in its structure).

are summarized in the SI (Table S6). A consistent trend in the brain-to-water ratio of PFAS concentrations was observed, in which PFAS with a higher number of carbons showed higher values for the C_b/C_w ratios for PFCAs and PFSAs. This trend aligned with trends observed for the C_b/C_s ratios in the human brain and serum analysis when comparing PFAS with different number of carbons although with much higher overall C_b/C_w ratios compared to the C_b/C_s values. This outcome highlighted the fact that the number of carbons of these PFAS is a significant factor that affects the partitioning of PFAS to the brain.

Perfluoroalkane sulfonamido acetic acids (FASAAs) show a different pattern. *N*-MeFOSAA, a chemical of the FOSAA group, exhibited a different trend in the C_b/C_s compared to PFCAs and PFSAs in human brain and serum analyses. Those results suggest that other physiochemical properties may be important for the presence of these compounds in the brain.

3.3. PFAS Entry into the Brain. The results show two important findings: (1) PFAS clearly penetrate the brain and (2) the total number of carbons of PFAS is a key factor in determining the brain-to-blood concentration ratios. While understanding the mechanism of PFAS entry into the brain is beyond the scope of the current study, our findings indicate that relative accumulation may be a function of the physical-chemical properties of the PFAS such as relative hydrophobicity, which is a function of carbon-chain length. The current understanding of how PFAS cross the blood−brain barrier is not clear, but several studies have suggested potential factors that might affect PFAS movement across the barrier, including interactions with transporters, proteins, and lipid content.[25−29]

Stevenson et al.[28] demonstrated that longer-chain PFAS have greater inhibitory effects on one of the efflux transporters located at the blood−brain barrier compared to shorter-chain PFAS. If shorter-chain PFAS are more effectively removed from the brain by efflux transporters due to their weaker inhibitory effects, the brain-to-serum concentration ratios would be expected to be lower for shorter-chain PFAS than those of longer-chain PFAS. This potential explanation aligns with the findings of the present study, where PFAS with fewer carbons exhibited lower brain-to-serum concentration ratios.

Several studies suggested that the affinity of charged molecules for phospholipids in biological membranes facilitates the bioaccumulation of PFAS.[30−32] The ion charge level of each PFAS might affect their distribution between serum and brain, as it can influence the extent of their partitioning into phospholipids within the brain tissue membrane. Furthermore, Dassuncao et al.[26] proposed that partitioning into phospholipids is a vital mechanism influential to the bioaccumulation of long-chain PFAS, particularly PFCAs with more than 10 carbon atoms. They found that the relative concentrations of PFAS in the brain increased with carbon-chain lengths.[26] Their findings suggest that longer-chain PFAS, which are more hydrophobic than shorter-chain PFAS, might be able to enter the brain by partitioning into the phospholipids in the brain tissue membrane. The higher brain-to-serum concentration ratios observed for PFAS with more carbons in the present study may be attributed to their greater hydrophobicity, which facilitates their partitioning into lipid-rich brain tissue.

Another proposed mechanism for PFAS accumulation in tissues is the interaction of PFAS with proteins, particularly serum albumin.[30−32] The ability of PFAS to bind to serum albumin could be a critical determinant of PFAS brain-to-serum concentration ratios. Although this mechanism has not been specifically suggested for the accumulation or distribution of PFAS in the brain, they could be significant influencers on our observations of brain-to-serum concentration ratios, given the high albumin content in serum.[26]

Another study has also highlighted the potential influence of membranes and albumin on PFAS partitioning into the brain.[33] To assess its potential relationship with the present study, we plotted measured brain-to-serum (C_b/C_s) and brain-to-water (C_b/C_w) concentration ratios of PFAS compounds against membrane-water distribution coefficients (DMW) and albumin-water distribution coefficients (DALBW) reported by Allendorf et al.[33] (Figures S1 and S2). A consistent positive

https://doi.org/10.1021/acs.est.4c09458
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relationship between both DMW and DALBW and the C_b/C_w values obtained from PFAS-spiked goat and sheep brain samples was observed. However, the relationship was less clear for the C_b/C_s values obtained from a human brain sample. This discrepancy may be attributed to the fact that DMW and DALBW trends align better with lower-carbon-chain PFAS (showing an increase with carbon-chain length) but become inconsistent for higher-carbon-chain PFAS. For higher-carbon-chain PFCAs, neither DALBW nor DMW consistently increases with carbon-chain length, whereas our measured C_b/C_s and C_b/C_w exhibit a clear increasing trend with an increase in the number of carbons. Notably, Allendorf et al.[33] acknowledged this inconsistency for higher-carbon-chain PFAS but did not provide a solid explanation. Overall, the trends that DALBW and DMW as well as C_b/C_s and C_b/C_w increase with the number of carbons suggest that the observed measurements are related to the partitioning of PFAS between serum and the brain rather than being influenced by kinetics or selective rejection of PFAS at the blood−brain barrier.

4. LIMITATIONS AND IMPLICATIONS

Although 18 PFAS compounds out of 43 PFAS analyzed were detected in at least one serum or brain sample, leading to the key finding that the number of carbons is a critical determinant for PFAS brain-to-serum concentration ratios, several PFAS were not detected or only detected in limited number of samples. Given the limited sample size in this study, further study with a larger sample size is required to fully understand the presence and accumulation of a wider range of PFAS in the human brain. The partition experiments in this study simply aimed to provide potential explanations for the observed brain-to-serum concentration ratios in the human sample analysis and did not aim to determine a partition coefficient. To estimate accurate partition coefficients of PFAS between water and brain, further experiments are required, which involve consideration of multiple factors, including chemical equilibrium between phases over time, binding affinities of PFAS to brain tissue components, the presence of dissolved lipids in the supernatant, and pH of the solution. In this study, LOQs were employed in place of method detection limits (MDLs) due to the unavailability of clean (PFAS-free) reference brain or serum samples, which are typically preferred for determining MDLs. We acknowledge that using instrumental LOQs instead of MDLs might introduce certain limitations as the LOQs were not established using a matrix identical to the analyzed samples. However, all of the usual criteria were used to ensure that the identification/quantification of analytes is correct (i.e., retention time, ion ratio, signal-to-noise >3). These measures enhance the confidence in the observed peaks and the validity of the results in this study.

This study provides evidence of the presence of PFAS in human brain tissue. The relative concentrations of PFAS in the brain, compared with those in the serum, increase with the number of carbons for PFCAs and PFSAs. There was no difference in PFAS concentrations between the two cortical brain regions analyzed, which is an important finding for our further understanding of PFAS distribution within the brain. Understanding brain-to-serum ratios of PFAS concentrations for each PFAS might enable us to use PFAS levels in the serum as proxies for brain exposure in combined-exposure biomonitoring and epidemiological studies. Moreover, additional research is needed to explore the impact of PFAS accumulation on brain health, particularly longer-chain PFAS, which is likely of greater concern.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.est.4c09458.

Additional information on chemicals and standards, PFAS extraction methods, LC-MS/MS analysis and quantification, QA/QC results, hemoglobin measurement results, partition experiment results, and comparison between the results of the present study and membrane/albumin-water distribution coefficients (PDF)

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Notes

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■ ACKNOWLEDGMENTS

M.S. is funded by The University of Queensland Research Training Scholarships and Minderoo Foundation. The Minderoo Centre−Plastics and Human Health gratefully acknowledges the support of the Minderoo Foundation, Australia. The Queensland Alliance for Environmental Health Sciences, The University of Queensland, gratefully acknowledges the financial support of Queensland Health, Australia, and the contribution of the Sydney Brain Bank and Neuroscience Research Australia for providing serum and brain samples. Tissues were received from the Sydney Brain Bank which is supported by Neuroscience Research Australia and a special gift in memory of Jim Raftos from the Shaw family. J.F.M. is the recipient of an ARC Australian Laureate Fellowship [project number FL200100028]. We acknowledge all of the patients and their families for serum and brain donation. The paired sample collections were supported by funding to G.H. from the National Health and Medical Research Council of Australia [Program Grant 132524; Investigator Grant 1176607], and we are grateful to the laboratory staff for blood and brain processing. Graphical abstract was created in https://BioRender.com.

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